

Change the title of Clause 167 (added by IEEE Std 802.3db-2022) as follows:

**167. Physical Medium Dependent (PMD) sublayer and medium, type
100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 800GBASE-VR8,
100GBASE-SR1, 200GBASE-SR2, ~~and 400GBASE-SR4~~, and 800GBASE-SR8**

167.1 Overview

Change the text of 167.1 as follows:

This clause specifies the 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 800GBASE-VR8, 100GBASE-SR1, 200GBASE-SR2, ~~and 400GBASE-SR4~~, and 800GBASE-SR8 PMDs together with the multimode fiber medium. The optical signals generated by these PMD types are modulated using a 4-level pulse amplitude modulation (PAM4) format. The PMD sublayers provide point-to-point 100, 200, ~~and 400~~, and 800 Gigabit Ethernet links over one, two, ~~or four~~, or eight pairs of multimode fiber, respectively.

When forming a complete Physical Layer, a PMD shall be connected to the appropriate PMA, as shown in Table 167-1, ~~or Table 167-2~~, or Table 167-2a, to the medium through the MDI and optionally with the management functions that may be accessible through the management interface defined in Clause 45, or equivalent. Figure 167-1 shows the relationship of the PMDs and MDIs (shown shaded) with other sublayers to the ISO/IEC Open System Interconnection (OSI) reference model. 100 Gigabit Ethernet is introduced in Clause 80 and the purpose of each PHY sublayer is summarized in 80.2. 200 Gigabit Ethernet and 400 Gigabit Ethernet are introduced in Clause 116 and the purpose of each PHY sublayer is summarized in 116.2. 800 Gigabit Ethernet is introduced in Clause 169 and the purpose of each PHY sublayer is summarized in 169.2.

100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, and 400GBASE-SR4 PHYs with the optional Energy-Efficient Ethernet (EEE) fast wake capability may enter the Low Power Idle (LPI) mode to conserve energy during periods of low link utilization (see Clause 78). The deep sleep mode of EEE is not supported. 800GBASE-VR8 and 800GBASE-SR8 PHYs do not support EEE capability.

The 100GBASE-VR1, 200GBASE-VR2, ~~and 400GBASE-VR4~~, and 800GBASE-VR8 sublayers provide point-to-point 100, 200, ~~and 400~~, and 800 Gigabit Ethernet links over one, two, ~~or four~~, or eight pairs of multimode fiber, respectively, up to at least 30 m on OM3, 50 m on OM4, and 50 m on OM5 (see 167.7). The 100GBASE-SR1, 200GBASE-SR2, ~~and 400GBASE-SR4~~, and 800GBASE-SR8 sublayers provide point-to-point 100, 200, ~~and 400~~, and 800 Gigabit Ethernet links over one, two, ~~or four~~, or eight pairs of multimode fiber, respectively, up to at least 60 m on OM3, 100 m on OM4, and 100 m on OM5 (see 167.7).

Further relevant information may be found in Clause 1 (terminology and conventions, references, definitions, and abbreviations) and Annex A (bibliography, referenced as [B1], [B2], etc.).

Insert new Table 167–2a after Table 167–2 as follows:

Table 167–2a—Physical Layer clauses associated with the 800GBASE-VR8 and 800GBASE-SR8 PMDs

Associated clause	800GBASE-VR8, 800GBASE-SR8	Status
170	RS	Required
170	800GMII ^a	Optional
171	800GMII Extender	Optional
172	800GBASE-R PCS	Required
173	800GBASE-R PMA	Required
120F	800GAUI-8 C2C	Optional
120G	800GAUI-8 C2M	Optional

^a The 800GMII is an optional interface. However, if the appropriate interface is not implemented, a conforming implementation behaves functionally as though the RS and 800GMII was present.

Replace Figure 167–1 with the following figure:

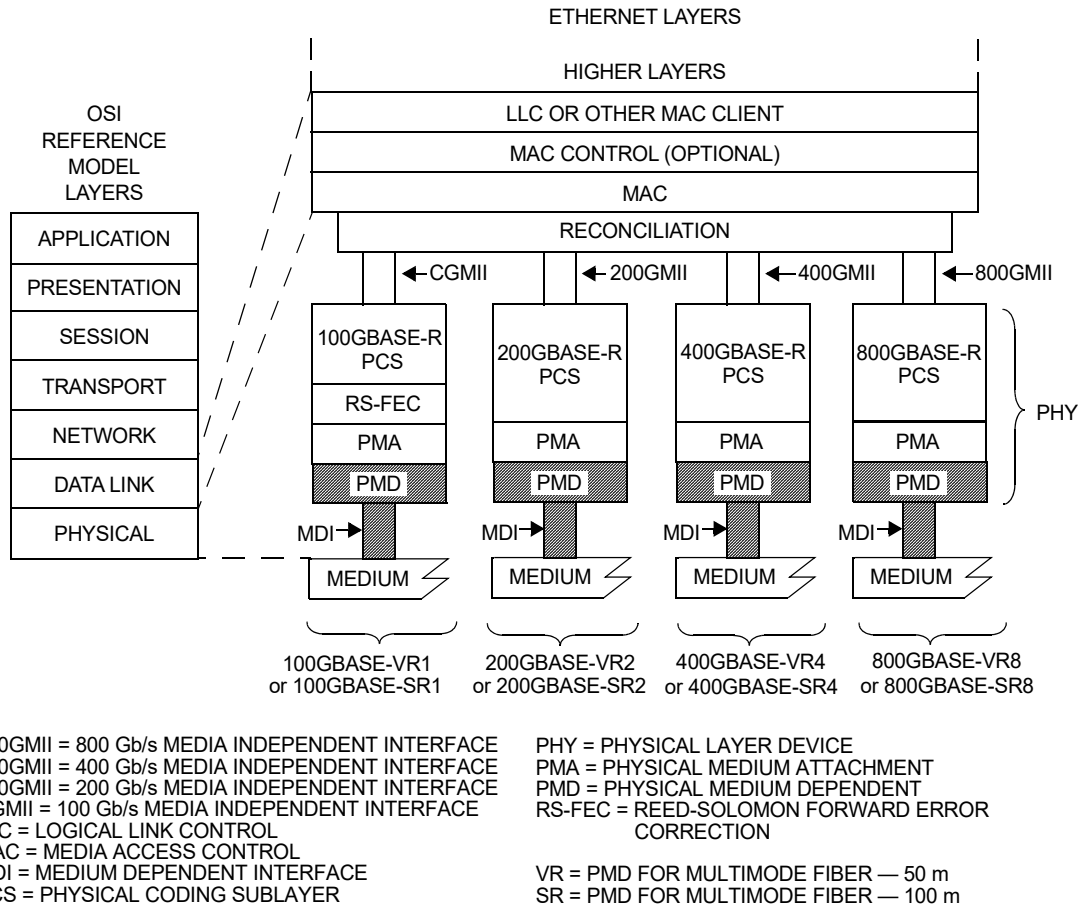


Figure 167–1—100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 800GBASE-VR8, 100GBASE-SR1, 200GBASE-SR2, 400GBASE-SR4, and 800GBASE-SR8 PMD relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

167.1.1 Bit error ratio

Insert a new paragraph at the end of 167.1.1 as follows:

For the 800GBASE-VR8 and 800GBASE-SR8 PMDs, the bit error ratio (BER) when processed by the PMA (Clause 173) shall be less than 2.4×10^{-4} provided that the error statistics are sufficiently random that this results in a frame loss ratio (see 1.4.344) of less than 3.4×10^{-12} for 64-octet frames with minimum interpacket gap when processed according to Clause 173 and then Clause 172. For a complete Physical Layer, the frame loss ratio may be degraded to 6.2×10^{-11} for 64-octet frames with minimum interpacket gap due to additional errors from the electrical interfaces. If the error statistics are not sufficiently random to meet this requirement, then the BER shall be less than that required to give a frame loss ratio of less than 3.4×10^{-12} for 64-octet frames with minimum interpacket gap.

167.2 Physical Medium Dependent (PMD) service interface

Change the first paragraph in 167.2 as follows:

This subclause specifies the services provided by the 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 800GBASE-VR8, 100GBASE-SR1, 200GBASE-SR2, and 400GBASE-SR4, and 800GBASE-SR8 PMDs. The service interfaces for these PMDs are described in an abstract manner and do not imply any particular implementation. The PMD service interface supports the exchange of encoded data between the PMA entity that resides just above the PMD, and the PMD entity. The PMD translates the encoded data to and from signals suitable for the specified medium.

Insert a new paragraph after the fourth paragraph in 167.2 as follows:

The 800GBASE-VR8 or 800GBASE-SR8 PMD service interface is an instance of the inter-sublayer service interface defined in 169.3, with eight parallel symbol streams ($n = 8$).

Insert a new paragraph after the eighth (now ninth) paragraph in 167.2 as follows:

The 800GBASE-VR8 or 800GBASE-SR8 PMD has eight parallel symbol streams, hence $i = 0$ to 7.

167.3 Delay and Skew

167.3.1 Delay constraints

Insert a new paragraph after the fourth paragraph in 167.3.1 as follows:

The sum of the transmit and receive delays at one end of the link contributed by the 800GBASE-VR8 or 800GBASE-SR8 PMD including 2 m of fiber in one direction shall be no more than 16 384 bit times (32 pause_quanta or 20.48 ns).

Change the last paragraph in 167.3.1 as follows:

Descriptions of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 80.4 for 100GBASE-VR1 and 100GBASE-SR1, ~~and in 116.4 and its references~~ for 200GBASE-VR2, 200GBASE-SR2, 400GBASE-VR4, and 400GBASE-SR4, and in 169.4 for 800GBASE-VR8 and 800GBASE-SR8.

167.3.2 Skew constraints

Change 167.3.2 as follows:

The Skew (relative delay) between the PCS or FEC lanes is kept within limits so that the information on the PCS or FEC lanes can be reassembled by the PCS or FEC. The Skew Variation also needs to be limited to ensure that a given PCS or FEC lane always traverses the same physical lane.

167.3.2.1 Skew constraints for 100GBASE-VR1, 100GBASE-SR1, 200GBASE-VR2, 200GBASE-SR2, 400GBASE-VR4, and 400GBASE-SR4

Skew and Skew Variation are defined in 80.5 and 116.5 and specified at the points SP1 to SP6 shown in Figure 80–8 and Figure 116–5.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 43 ns and the Skew Variation at SP2 is limited to 400 ps. For 100GBASE-VR1

and 100GBASE-SR1, since the signal at the PMD service interface represents a serial bit stream, there is no Skew Variation at this point.

The Skew at SP3 (the transmitter MDI) shall be less than 54 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 134 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 145 ns and the Skew Variation at SP5 shall be less than 3.6 ns. For 100GBASE-VR1 and 100GBASE-SR1, since the signal at the PMD service interface represents a serial bit stream, there is no Skew Variation at this point.

For more information on Skew and Skew Variation see 80.5 and 116.5.

167.3.2.2 Skew constraints for 800GBASE-VR8 and 800GBASE-SR8

Skew and Skew Variation are defined in 169.5 and specified at the points SP1 to SP6 shown in Figure 169-4.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 25 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 36 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 116 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 127 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation see 169.5.

167.5 PMD functional specifications

167.5.1 PMD block diagram

Change the first paragraph in 167.5.1 as follows:

The PMD block diagram for 400GBASE-VR4 or 400GBASE-SR4 is shown in Figure 167-2. The block diagrams for ~~200GBASE-VR2~~ 100GBASE-VR1 and ~~200GBASE-SR2~~ 100GBASE-SR1 are equivalent to Figure 167-2, but for ~~two lanes~~ one lane per direction. The block diagrams for ~~100GBASE-VR1~~ 200GBASE-VR2 and ~~100GBASE-SR1~~ 200GBASE-SR2 are equivalent to Figure 167-2, but for ~~one lane~~ two lanes per direction. The block diagrams for 800GBASE-VR8 and 800GBASE-SR8 are equivalent to Figure 167-2, but for eight lanes per direction.

167.5.2 PMD transmit function

Change 167.5.2 as follows:

The PMD Transmit function shall convert the one, two, ~~or four, or eight~~ symbol streams requested by the PMD service interface messages PMD:IS_UNITDATA_*i*.request into one, two, ~~or four, or eight~~ separate optical signals. The 100GBASE-VR1 and 100GBASE-SR1 PMDs have a single symbol stream, hence *i* = 0. The 200GBASE-VR2 and 200GBASE-SR2 PMDs have two parallel symbol streams, hence *i* = 0 to 1. The 400GBASE-VR4 and 400GBASE-SR4 PMDs have four parallel symbol streams, hence *i* = 0 to 3. The 800GBASE-VR8 and 800GBASE-SR8 PMDs have eight parallel symbol streams, hence *i* = 0 to 7. Each optical signal shall then be delivered to the MDI, which contains one, two, ~~or four, or eight~~ parallel light paths for transmit, according to the transmit optical specifications in this clause. The four optical power levels in each signal in order from lowest to highest shall correspond to tx_symbol values zero, one, two, and three, respectively.

167.5.3 PMD receive function

Change 167.5.3 as follows:

The PMD Receive function shall convert the one, two, ~~or four, or eight~~ parallel optical signals received from the MDI into one, two, ~~or four, or eight~~ separate symbol streams for delivery to the PMD service interface using the messages PMD:IS_UNITDATA_0.indication to PMD:IS_UNITDATA_*i*.indication, all according to the receive optical specifications in this clause. The four optical power levels in each signal in order from lowest to highest shall correspond to rx_symbol values zero, one, two, and three, respectively.

167.5.4 PMD global signal detect function

Change the second paragraph in 167.5.4 as follows:

SIGNAL_DETECT shall be a global indicator of the presence of optical signals on all lanes. The value of the SIGNAL_DETECT parameter shall be generated according to the conditions defined in Table 167–5. The PMD receiver is not required to verify whether a compliant 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 800GBASE-VR8, 100GBASE-SR1, 200GBASE-SR2, ~~or 400GBASE-SR4, or 800GBASE-SR8~~ signal is being received. This standard imposes no response time requirements on the generation of the SIGNAL_DETECT parameter.

Change Table 167–5 as follows:

Table 167–5—SIGNAL_DETECT value definition

Receive conditions	SIGNAL_DETECT value
For any lane; Average optical power at TP3 ≤ -30 dBm	FAIL
For all lanes; [(Optical power at TP3 $\geq$ average receive power, each lane (min) in Table 167–8) AND (compliant 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 800GBASE-VR8, 100GBASE-SR1, 200GBASE-SR2, or 400GBASE-SR4, or 800GBASE-SR8 signal input)]	OK
All other conditions	Unspecified

167.6 Lane assignments

Change 167.6 as follows:

There are no lane assignments (within a group of transmit or receive lanes) for 200GBASE-VR2, 400GBASE-VR4, 800GBASE-VR8, 200GBASE-SR2, ~~or 400GBASE-SR4, or 800GBASE-SR8~~. While it is expected that a PMD will map electrical lane i to optical lane i and vice versa, there is no need to define the physical ordering of the lanes, as the PCS sublayer is capable of receiving the lanes in any arrangement. The positioning of transmit and receive lanes at the MDI is specified in [167.10.3.1](#) and [167.10.3.1a](#).

Change the title and text of 167.7 as follows:

167.7 PMD to MDI optical specifications for 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 800GBASE-VR8, 100GBASE-SR1, 200GBASE-SR2, ~~and 400GBASE-SR4, and 800GBASE-SR8~~

The operating ranges for the 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 800GBASE-VR8, 100GBASE-SR1, 200GBASE-SR2, ~~and 400GBASE-SR4, and 800GBASE-SR8~~ PMDs are defined in Table 167–6. A compliant PMD operates on 50/125 μm multimode fibers, type A1a.2 (OM3), type A1a.3 (OM4), or type A1a.4 (OM5), according to the specifications defined in Table 167–14. A PMD that exceeds the operating range requirement while meeting all other optical specifications is considered compliant (e.g., a 100GBASE-SR1 PMD operating at 120 m meets the operating range requirement of 0.5 m to 100 m).

Change Table 167–6 as follows:

Table 167–6—Operating range

PMD type	Required operating range ^a
100GBASE-VR1	0.5 m to 30 m for OM3
200GBASE-VR2	
400GBASE-VR4	0.5 m to 50 m for OM4
<u>800GBASE-VR8</u>	0.5 m to 50 m for OM5
100GBASE-SR1	0.5 m to 60 m for OM3
200GBASE-SR2	
400GBASE-SR4	0.5 m to 100 m for OM4
<u>800GBASE-SR8</u>	0.5 m to 100 m for OM5

^a The RS-FEC correction function may not be bypassed for any operating distance.

167.7.1 Transmitter optical specifications

Change the first paragraph in 167.7.1 as follows:

Each lane of a 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 800GBASE-VR8, 100GBASE-SR1, 200GBASE-SR2, ~~and 400GBASE-SR4, and 800GBASE-SR8~~ transmitter shall meet the specifications in Table 167–7 per the definitions in 167.8.

Change Table 167–7 as follows:

Table 167–7—Transmit characteristics

Description	100GBASE-VR1 200GBASE-VR2 400GBASE-VR4 800GBASE-VR8	100GBASE-SR1 200GBASE-SR2 400GBASE-SR4 800GBASE-SR8	Unit
Signaling rate, each lane (range) <u>Other PMDs</u> <u>800GBASE-VR8, 800GBASE-SR8 PMDs</u>	53.125 ± 100 ppm <u>53.125 ± 50 ppm</u>		GBd <u>GBd</u>
Modulation format	PAM4		—
Center wavelength (range)	842 to 948	844 to 863	nm
RMS spectral width ^a (max)	0.65	0.6	nm
Average launch power, each lane (max)	4		dBm
Average launch power, each lane (min)	−4.6		dBm
Outer Optical Modulation Amplitude (OMA _{outer}), each lane (max)	3.5		dBm
Outer Optical Modulation Amplitude (OMA _{outer}), each lane (min) For max (TECQ, TDECQ) ≤ 1.8 dB For 1.8 < max (TECQ, TDECQ) ≤ 4.4 dB	−2.6 −4.4 + max (TECQ, TDECQ)		dBm dBm
Transmitter and dispersion eye closure for PAM4 (TDECQ), each lane (max)	4.4	4.4	dB
Transmitter eye closure for PAM4 (TECQ), each lane (max)	4.4		dB
Overshoot/undershoot (max)	29		%
Transmitter power excursion, each lane (max)	2.3		dBm
Extinction ratio, each lane (min)	2.5		dB
Transmitter transition time, each lane (max)	17		ps
Average launch power of OFF transmitter, each lane (max)	−30		dBm
RIN ₁₄ OMA (max)	−132		dB/Hz
Optical return loss tolerance (max)	14		dB
Encircled flux ^b	≥ 86% at 19 μm ≤ 30% at 4.5 μm		—

^a RMS spectral width is the standard deviation of the spectrum.

^b If measured into type A1a.2 or type A1a.3, or A1a.4, 50 μm fiber, in accordance with IEC 61280-1-4.

167.7.2 Receiver optical specifications

Change the first paragraph of 167.7.2 as follows:

Each lane of a 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 800GBASE-VR8, 100GBASE-SR1, 200GBASE-SR2, ~~and 400GBASE-SR4,~~ and 800GBASE-SR8 receiver shall meet the specifications in Table 167–8 per the definitions in 167.8.

Change Table 167–8 as follows:

Table 167–8—Receive characteristics

Description	100GBASE-VR1 200GBASE-VR2 400GBASE-VR4 <u>800GBASE-VR8</u>	100GBASE-SR1 200GBASE-SR2 400GBASE-SR4 <u>800GBASE-SR8</u>	Unit
Signaling rate, each lane (range) <u>Other PMDs</u> <u>800GBASE-VR8, 800GBASE-SR8 PMDs</u>	53.125 ± 100 ppm <u>53.125 ± 50 ppm</u>		GBd <u>GBd</u>
Modulation format	PAM4		—
Center wavelength (range)	842 to 948		nm
Damage threshold ^a (min)	5		dBm
Average receive power, each lane (max)	4		dBm
Average receive power, each lane ^b (min)	–6.3	–6.4	dBm
Receive power, each lane (OMA _{outer}) (max)	3.5		dBm
Receiver reflectance (max)	–15		dB
Receiver sensitivity (OMA _{outer}) (max) For TECQ ≤ 1.8 dB For 1.8 < TECQ ≤ 4.4 dB	–4.4 –6.2 + TECQ	–4.6 –6.4 + TECQ	dBm dBm
Stressed receiver sensitivity (OMA _{outer}) ^c (max)	–1.8	–2	dBm
Conditions of stressed receiver sensitivity test: ^d			
Stressed eye closure for PAM4 (SECQ), lane under test	4.4		dB
OMA _{outer} of each aggressor lane	3.5		dBm

^a The receiver shall be able to tolerate, without damage, continuous exposure to an optical input signal having this average power level on one lane. The receiver does not have to operate correctly at this input power.

^b Average receive power, each lane (min) is not the principal indicator of signal strength. A received power below this value cannot be compliant; however, a value above this does not ensure compliance.

^c Measured with conformance test signal at TP3 (see 167.8.14) for the BER specified in 167.1.1.

^d These test conditions are for measuring stressed receiver sensitivity. They are not characteristics of the receiver.

167.7.3 Illustrative link power budget

Change the first paragraph in 167.7.3 as follows:

An illustrative power budget and penalties for 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 800GBASE-VR8, 100GBASE-SR1, 200GBASE-SR2, ~~and 400GBASE-SR4~~, and 800GBASE-SR8 channels are shown in Table 167–9.

Change Table 167–9 as follows:

Table 167–9—Illustrative link power-budget budgets

Parameter	100GBASE-VR1 200GBASE-VR2 400GBASE-VR4 800GBASE-VR8			100GBASE-SR1 200GBASE-SR2 400GBASE-SR4 800GBASE-SR8			Unit
	OM3	OM4	OM5	OM3	OM4	OM5	
Effective modal bandwidth at 850 nm ^a	2000	4700		2000	4700		MHz · km
Power budget (for max TDECQ)	6.2			6.4			dB
Operating distance	0.5 to 30	0.5 to 50		0.5 to 60	0.5 to 100		m
Channel insertion loss ^b	1.6	1.7		1.7	1.8		dB
Allocation for penalties ^c (for max TDECQ)	4.5			4.6			dB
Additional insertion loss allowed	0.1	0		0.1	0		dB

^a Per IEC 60793-2-10, see Table 167–15 for information on effective modal bandwidth at other wavelengths.

^b The channel insertion loss is calculated using the maximum distance specified in Table 167–6 and cabled optical fiber attenuation of 3 dB/km at 850 nm plus an allocation for connection and splice loss given in 167.10.2.2.1.

^c Link penalties are used for link budget calculations. They are not requirements and are not meant to be tested.

167.8 Definition of optical parameters and measurement methods

167.8.1 Test patterns for optical parameters

Change Table 167–10 as follows:

Table 167–10—Test patterns

Pattern	Pattern description	Defined in
Square wave	Square wave (8 threes, 8 zeros)	120.5.11.2.4
3	PRBS31Q	120.5.11.2.2
4	PRBS13Q	120.5.11.2.1
5	Scrambled idle encoded by RS-FEC	82.2.11 and 91, or 119.2.4.9, or 172.2.4.11
6	SSPRQ	120.5.11.2.3

Change Table 167–11 as follows:

Table 167–11—Test-pattern definitions and related subclauses

Parameter	Pattern	Related subclause
Wavelength, spectral width	3, 4, 5, 6, or valid 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, or <u>400GBASE-SR4-100GBASE-R,</u> <u>200GBASE-R, 400GBASE-R, or</u> <u>800GBASE-R signal</u>	167.8.3
Average optical power	3, 4, 5, 6, or valid 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, or <u>400GBASE-SR4-100GBASE-R,</u> <u>200GBASE-R, 400GBASE-R, or</u> <u>800GBASE-R signal</u>	167.8.4
Outer Optical Modulation Amplitude (OMA _{outer})	4 or 6	167.8.5
Transmitter and dispersion eye closure for PAM4 (TDECQ)	6	167.8.6
Transmitter eye closure for PAM4 (TECQ)	6	167.8.7
Overshoot/undershoot	6	167.8.8
Transmitter power excursion	6	167.8.9
Extinction ratio	4 or 6	167.8.10
Transmitter transition time	Square wave or 6	167.8.11
RIN ₁₄ OMA	Square wave	167.8.12
Receiver sensitivity	3 or 5	167.8.13
Stressed receiver sensitivity	3, 5, or valid 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, or <u>400GBASE-SR4-100GBASE-R,</u> <u>200GBASE-R, 400GBASE-R, or</u> <u>800GBASE-R signal</u>	167.8.14
Stressed eye closure (SEC), calibration	6	167.8.14

167.8.6 Transmitter and dispersion eye closure for PAM4 (TDECQ).

Change Table 167–12 as follows:

Table 167–12—The 3 dB bandwidth f_A of the fiber emulation filter for TDECQ measurement

PMD type	Center wavelength (range) (nm)	3 dB bandwidth f_A (GHz)
100GBASE-VR1 200GBASE-VR2 400GBASE-VR4 <u>800GBASE-VR8</u>	842 to 868	33.6
	842 to 888	29.6
	842 to 918	24.5
	842 to 948	20.7
100GBASE-SR1 200GBASE-SR2 400GBASE-SR4 <u>800GBASE-SR8</u>	844 to 863	18.0

167.8.13 Receiver sensitivity

Change the second paragraph in 167.8.13 as follows:

The conformance test signal at TP3 meets the requirements for a 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 800GBASE-VR8, 100GBASE-SR1, 200GBASE-SR2, ~~or 400GBASE-SR4~~, or 800GBASE-SR8 transmitter followed by an attenuator. For multi-lane testing considerations, see 167.8.2. The BER requirements are given in 167.1.1.

167.9 Safety, installation, environment, and labeling

167.9.2 Laser safety

Change the first paragraph in 167.9.2 as follows:

100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 800GBASE-VR8, 100GBASE-SR1, 200GBASE-SR2, ~~and 400GBASE-SR4~~, and 800GBASE-SR8 optical transceivers shall conform to Hazard Level 1M laser requirements as defined in IEC 60825-1 and IEC 60825-2, under any condition of operation. This includes single fault conditions whether coupled into a fiber or out of an open bore.

167.9.4 Environment

Change the first paragraph in 167.9.4 as follows:

Normative specifications in this clause shall be met by a system integrating a 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 800GBASE-VR8, 100GBASE-SR1, 200GBASE-SR2, ~~or 400GBASE-SR4~~, or 800GBASE-SR8 PMD over the life of the product while the product operates within the manufacturer's range of environmental, power, and other specifications.

167.9.5 Electromagnetic emission

Change 167.9.5 as follows:

A system integrating a 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 800GBASE-VR8, 100GBASE-SR1, 200GBASE-SR2, or 400GBASE-SR4, or 800GBASE-SR8 PMD shall comply with applicable local and national codes for the limitation of electromagnetic interference.

167.9.7 PMD labeling requirements

Change the first paragraph in 167.9.7 as follows:

It is recommended that each PHY (and supporting documentation) be labeled in a manner visible to the user, with at least the applicable safety warnings and the applicable port type designation (e.g., 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 800GBASE-VR8, 100GBASE-SR1, 200GBASE-SR2, or 400GBASE-SR4, or 800GBASE-SR8). It is recommended that each PHY with an angled fiber interface indicate that it uses an angled MDI.

167.10 Fiber optic cabling model

Change 167.10 as follows:

The fiber optic cabling (channel) contains one optical fiber for each direction to support 100GBASE-VR1 or 100GBASE-SR1, two optical fibers for each direction to support 200GBASE-VR2 or 200GBASE-SR2, ~~or~~ four optical fibers for each direction to support 400GBASE-VR4 or 400GBASE-SR4, or eight optical fibers for each direction to support 800GBASE-VR8 or 800GBASE-SR8. The fiber optic cabling interconnects the transmitters at the MDI on one end of the channel to the receivers at the MDI on the other end of the channel. As defined in 167.10.3, the optical lanes appear in defined locations at the MDI but the locations are not assigned specific lane numbers within this standard because any transmitter lane may be connected to any receiver lane.

167.10.1 Fiber optic cabling model

Change Table 167–14 as follows:

Table 167–14—Fiber optic cabling (channel) characteristics

Description	100GBASE-VR1 200GBASE-VR2 400GBASE-VR4 <u>800GBASE-VR8</u>			100GBASE-SR1 200GBASE-SR2 400GBASE-SR4 <u>800GBASE-SR8</u>			Unit
	OM3	OM4	OM5	OM3	OM4	OM5	
Operating distance (max)	30	50		60	100		m
Cabling Skew (max) ^a	79						ns
Cabling Skew Variation ^{a, b} (max)	2.4						ns
Channel insertion loss ^c (max)	1.6	1.7		1.7	1.8		dB
Channel insertion loss (min)	0			0			dB

^a Applies only to 200GBASE-VR2, 400GBASE-VR4, 800GBASE-VR8, 200GBASE-SR2, and 400GBASE-SR4, and 800GBASE-SR8.

^b An additional 400 ps of Skew Variation could be caused by wavelength changes, which are attributable to the transmitter not the channel.

^c These channel insertion loss values include cable loss plus 1.5 dB allocated for connection and splice loss, over the wavelength range 842 nm to 948 nm for VR and over the wavelength range 844 nm to 863 nm for SR.

167.10.2.1 Optical fiber cable

Change Table 167–15 as follows:

Table 167–15—Optical fiber and cable characteristics

Description	OM3 ^a	OM4 ^b	OM5 ^c	Unit
Nominal core diameter	50			μm
Nominal fiber specification wavelength	850		850 and 953	nm
Effective modal bandwidth at 850 nm (min) ^d	2000	4700		MHz · km
Effective modal bandwidth at 953 nm (min)	Not Specified ^e		2470 ^e	MHz · km
Cabled optical fiber attenuation (max)	3.0			dB/km
Zero dispersion wavelength (λ_0) ^f	$1297 \leq \lambda_0 \leq 1328$			nm
Chromatic dispersion slope (max) (S_0) ^f	$-412/(840(1 - (\lambda_0/840)^4))$			ps/nm ² km

^a IEC 60793-2-10 type A1a.2.

^b IEC 60793-2-10 type A1a.3.

^c IEC 60793-2-10 type A1a.4.

^d When measured with the launch conditions specified in Table 167–7.

^e Guidance is provided for effective modal bandwidth at all wavelengths in the 840 nm to 953 nm range in IEC 60793-2-10.

^f These limits are consistent with IEC 60793-2-10/AMD1 ED7. For OM5, they are the same as previous versions of IEC 60793-2-10. OM3 and OM4 fibers compliant to previous versions of IEC 60793-2-10 are considered compliant for 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 800GBASE-VR8, 100GBASE-SR1, 200GBASE-SR2, ~~and 400GBASE-SR4~~, and 800GBASE-SR8 Physical Layer types.

167.10.3 Medium Dependent Interface (MDI)

Change 167.10.3 as follows:

The ~~100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, or 400GBASE-SR4~~ PMD is coupled to the fiber optic cabling at the MDI. The MDI is the interface between the PMD and the “fiber optic cabling” (as shown in [Figure 167–6](#)). Examples of an MDI include the following:

- PMD with a connectorized fiber pigtail plugged into an adapter;
- PMD receptacle.

NOTE—Transmitter compliance testing is performed at TP2 as defined in 167.5.1, not at the MDI.

Insert new subclause 167.10.3.1a including Figure 167–8a as follows:

167.10.3.1a Optical lane assignments for 800GBASE-VR8 and 800GBASE-SR8

The eight transmit and eight receive optical lanes of 800GBASE-VR8 or 800GBASE-SR8 shall occupy the positions depicted in Figure 167–8a when looking into the MDI receptacle with the connector keyway feature on top.

The interface contains 16 active lanes within 16 total positions. The transmit optical lanes occupy the leftmost eight positions. The receive optical lanes occupy the rightmost eight positions.

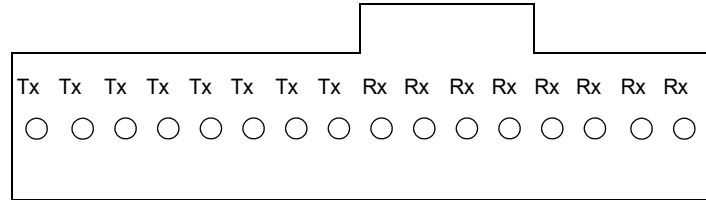


Figure 167–8a—Optical lane assignments for 800GBASE-VR8 and 800GBASE-SR8

Insert new subclause 167.10.3.4 after 167.10.3.3 as follows:

167.10.3.4 MDI requirements for 800GBASE-VR8 and 800GBASE-SR8

The MDI shall optically mate with the compatible plug on the optical fiber cabling. If the MDI is constructed with a connectorized fiber pigtail into an adapter, the connectorized pigtail shall meet the dimensional specifications of designation FOCIS 18 P-1x16-1-8-1-2-1 and the adapter shall meet the dimensional specifications of designation FOCIS 18 A-1-0 as defined in ANSI/TIA-604-18-A. If the MDI is constructed with a receptacle, it shall meet the dimensional specifications of designation FOCIS 18 R-1x16-1-8-1-2-2, as defined in ANSI/TIA-604-18-A. The plug terminating the optical fiber cabling that interconnects to either MDI configuration shall meet the dimensional specifications of designation FOCIS 18 P-1x16-1-8-2-2-1, as defined in ANSI/TIA-604-18-A. The MPO-16 female plug connector and MDI are structurally similar to those depicted in [Figure 167–10](#), but with an angled end facet, 16 fibers, an offset keyway, and different pin diameters and locations. The MDI connection shall meet the interface performance specifications of IEC 63267-1 for performance grade Bm/1m.

Change the title of 167.11 as follows:

**167.11 Protocol implementation conformance statement (PICS) proforma for
Clause 167, Physical Medium Dependent (PMD) sublayer and medium, type
100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 800GBASE-VR8,
100GBASE-SR1, 200GBASE-SR2, ~~and 400GBASE-SR4~~, and 800GBASE-SR8¹⁰**

167.11.1 Introduction

Change the first paragraph in 167.11.1 as follows:

The supplier of a protocol implementation that is claimed to conform to Clause 167, Physical Medium Dependent (PMD) sublayer and medium, type 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 800GBASE-VR8, 100GBASE-SR1, 200GBASE-SR2, ~~and 400GBASE-SR4~~, and 800GBASE-SR8, shall complete the following protocol implementation conformance statement (PICS) proforma.

167.11.2 Identification

167.11.2.2 Protocol summary

Change the table in 167.11.2.2 as follows:

Identification of protocol standard	IEEE Std 802.3df-2024, Clause 167, Physical Medium Dependent (PMD) sublayer and medium, type 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, <u>800GBASE-VR8</u> , 100GBASE-SR1, 200GBASE-SR2, and 400GBASE-SR4 , <u>and 800GBASE-SR8</u>
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3df-2024.)	

Date of Statement	
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¹⁰Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

167.11.3 Major capabilities/options

Change the table in 167.11.3 as follows (some unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
...					
*SR4	400GBASE-SR4 PMD	167.7	Device supports requirements for 400GBASE-SR4 PHY	O	Yes [] No []
*VR8	<u>800GBASE-VR8 PMD</u>	<u>167.7</u>	<u>Device supports requirements for 800GBASE-VR8 PHY</u>	<u>O</u>	<u>Yes []</u> <u>No []</u>
*SR8	<u>800GBASE-SR8 PMD</u>	<u>167.7</u>	<u>Device supports requirements for 800GBASE-SR8 PHY</u>	<u>O</u>	<u>Yes []</u> <u>No []</u>
*INS	Installation / cable	167.10.1	Items marked with INS include installation practices and cable specifications not applicable to a PHY manufacturer	O	Yes [] No []
...					

Change the title of 167.11.4 as follows:

167.11.4 PICS proforma tables for Physical Medium Dependent (PMD) sublayer and medium, type 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 800GBASE-VR8, 100GBASE-SR1, 200GBASE-SR2, ~~and~~ 400GBASE-SR4, and 800GBASE-SR8

167.11.4.1 PMD functional specifications

Change the row for item F1 in the table in 167.11.4.1 as follows (unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
F1	Compatible with 100GBASE-R, 200GBASE-R, or 400GBASE-R, <u>or 800GBASE-R</u> PCS and PMA	167.1		M	Yes []
...					

167.11.4.6 Characteristics of the fiber optic cabling and MDI

Change the table in 167.11.4.6 as follows (unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
...					
OC4	MDI layout for 400GBASE-VR2 and 400GBASE-SR2	167.10.3.1	Optical lane assignments per Figure 167-7	(VR2 or SR2) VR2+SR2:M	Yes [] N/A []
OC5	MDI layout for 400GBASE-VR4 and 400GBASE-SR4	167.10.3.1	Optical lane assignments per Figure 167-8	(VR4 or SR4) VR4+SR4:M	Yes [] N/A []
OC5a	MDI layout for 800GBASE-VR8 and 800GBASE-SR8	167.10.3.1a	Optical lane assignments per Figure 167-8a	VR8+SR8:M	Yes [] N/A []
OC6	MDI mating, 100GBASE-VR1 and 100GBASE-SR1, with duplex optical fiber connector	167.10.3.2	MDI optically mates with plug on the cabling, performance grade Bm/2m	(VR1 or SR1) VR1+SR1:M	Yes [] N/A []
OC7	MDI requirements for 100GBASE-VR1 and 100GBASE-SR1, with duplex optical fiber connector	167.10.3.2	Per IEC 61753-1 and IEC 61753-022-2	(VR1 or SR1) and INS:M INS* (VR1+SR1):M	Yes [] N/A []
OC8	MDI mating, with multifiber connector	167.10.3.3	MDI optically mates with plug on the cabling, performance grade Bm/2m	(VR1, SR1, VR2, SR2, VR4, or SR4):AFI:M !AFI* (VR1+SR1 +VR2+SR2 +VR4+SR4):M	Yes [] N/A []
OC9	MDI mating, with multifiber connector	167.10.3.3	MDI optically mates with plug on the cabling, performance grade Bm/1m	(VR1, SR1, VR2, SR2, VR4, or SR4):AFI:M AFI* (VR1+SR1 +VR2+SR2 +VR4+SR4):M	Yes [] N/A []
OC10	MDI dimensions, with multifiber connector	167.10.3.3	Per IEC 61754-7-1 interface 7-1-3 or interface 7-1-10	(VR1, SR1, VR2, SR2, VR4, or SR4):AFI:M !AFI* (VR1+SR1 +VR2+SR2 +VR4+SR4):M	Yes [] N/A []

Item	Feature	Subclause	Value/Comment	Status	Support
OC11	MDI dimensions, with multifiber connector	167.10.3.3	Per IEC 61754-7-1 interface 7-1-3 or interface 7-1-9	(VR1, SR1, VR2, SR2, VR4, or SR4):AFI:M AFI* (VR1+SR1+VR2+SR2+VR4+SR4):M	Yes [] N/A []
OC12	Cabling connector dimensions, with multifiber connector	167.10.3.3	Per IEC 61754-7-1 interface 7-1-4	INS and (VR1, SR1, VR2, SR2, VR4, or SR4):AFI:M INS*AFI* (VR1+SR1+VR2+SR2+VR4+SR4):M	Yes [] N/A []
OC13	Cabling connector dimensions, with multifiber connector	167.10.3.3	Per IEC 61754-7-1 interface 7-1-1	INS and (VR1, SR1, VR2, SR2, VR4, or SR4):AFI:M INS*AFI* (VR1+SR1+VR2+SR2+VR4+SR4):M	Yes [] N/A []
OC14	MDI requirements, with multifiber connector	167.10.3.3	Per IEC 61753-1 and IEC 61753-022-2, performance grade Bm/2m	INS and (VR1, SR1, VR2, SR2, VR4, or SR4):AFI:M INS*AFI* (VR1+SR1+VR2+SR2+VR4+SR4):M	Yes [] N/A []
OC15	MDI requirements, with multifiber connector	167.10.3.3	Per IEC 63267-1, performance grade Bm/1m	INS and (VR1, SR1, VR2, SR2, VR4, or SR4):AFI:M INS*AFI* (VR1+SR1+VR2+SR2+VR4+SR4):M	Yes [] N/A []
OC16	<u>MDI mating, with multifiber connector</u>	167.10.3.4	<u>MDI optically mates with plug on the cabling, performance grade Bm/1m</u>	VR8+SR8:M	Yes [] N/A []
OC17	<u>MDI dimensions, with multifiber connector</u>	167.10.3.4	<u>Per ANSI/TIA-604-18-A designation FOCIS 18 P-1x16-1-8-1-2-1 and designation FOCIS 18 A-1-0 or designation FOCIS 18 R-1x16-1-8-1-2-2</u>	VR8+SR8:M	Yes [] N/A []
OC18	<u>Cabling connector dimensions, with multifiber connector</u>	167.10.3.4	<u>Per ANSI/TIA-604-18-A designation FOCIS 18 P-1x16-1-8-2-2-1</u>	INS* (VR8+SR8):M	Yes [] N/A []
OC19	<u>MDI requirements, with multifiber connector</u>	167.10.3.4	<u>Per IEC 63267-1, performance grade Bm/1m</u>	INS* (VR8+SR8):M	Yes [] N/A []